NYC Vehicle Fatalities: A Borough-Specific Analysis for Targeted Safety Interventions

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Introduction and Objective

New York City's road safety challenges vary significantly by borough, making a "one-size-fits-all" strategy ineffective. This project analyzes fatal crash data to identify unique, borough-specific causes. The objective is to provide targeted, data-driven recommendations to improve safety for all, especially vulnerable road users.

Data & Methods

This analysis uses the NYC OpenData "Motor Vehicle Collisions" dataset (2014-2022), isolating all incidents with at least one fatality. Borough-level trends and contributing factors were analyzed and visualized using bar charts (Fig. 1), grouped bar charts (Fig. 2), and a geospatial heatmap (Fig. 3).

Key Findings & Analysis

Brooklyn and Queens account for over 60% of all traffic fatalities (Fig. 1). Each borough has a unique "crash signature" (Fig. 2), from intersection violations in Queens to unsafe speed in Staten Island. A major data gap exists due to "Unspecified" causes. Crash density is highest in Manhattan (Fig. 3).

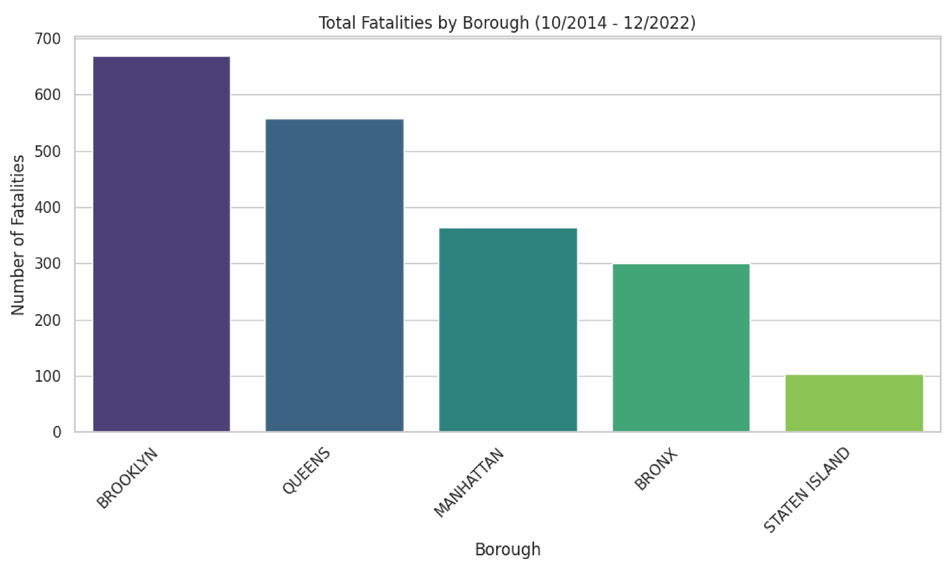

Figure 1. Total Fatalities by Borough (Bar Chart)

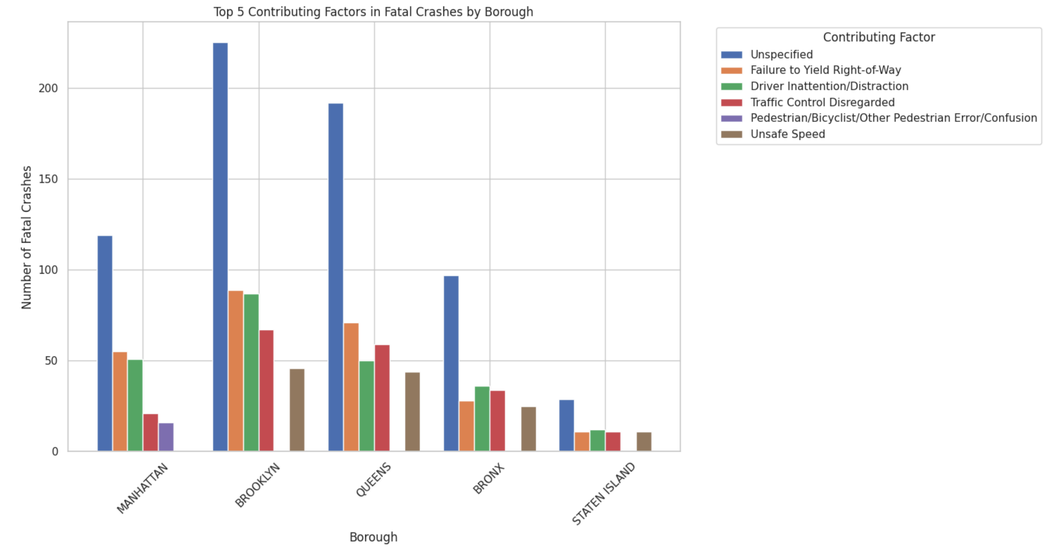

Figure 2. Top Contributing Factors in Fatal Crashes by Borough (Grouped Bar Chart)

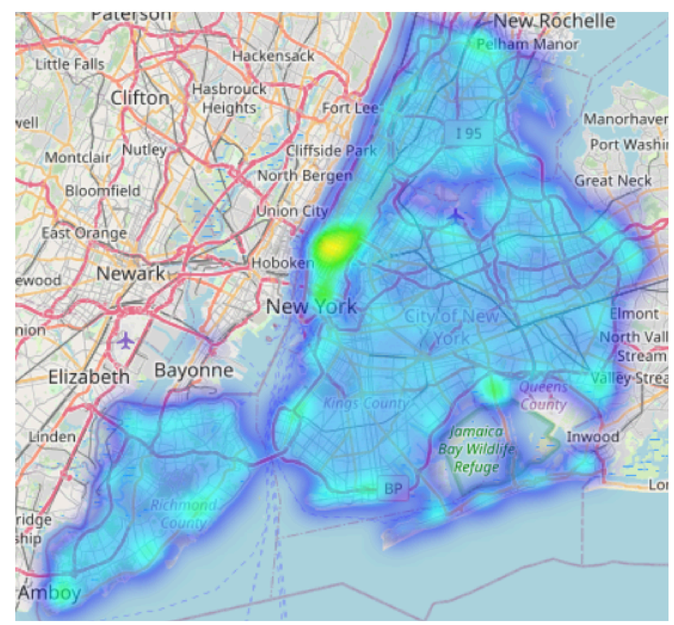

Figure 3. Geospatial Heatmap of Crash Density

Conclusions & Recommendations

A targeted, borough-specific strategy is essential. Interventions must align with local data: focus on intersection safety in Queens, distracted driving in Brooklyn, vulnerable road users in Manhattan, and speed reduction in Staten Island. Improving on-scene data collection to reduce "Unspecified" causes is a critical city-wide priority.

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